Butterworth Filter

A Complete Study Guide — From Basics to Full Design

How to Use These Notes

These notes are written for beginners. Each concept builds on the previous one. You do not need prior filter design experience — only basic knowledge of complex numbers and algebra. Every formula is derived step by step, and every claim is explained in plain language before the math is shown. Diagrams are provided for every key concept.

Suggested reading order: Read straight through from Section 1. Do not skip sections — each one prepares you for the next.

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1 What is a Filter and Why Do We Need One?

1.1 The Problem Filters Solve

In the real world, signals are almost never "clean." A microphone picks up both your voice *and* background hiss. A sensor records both the useful measurement *and* electrical noise. A radio antenna captures your station *and* dozens of others. We need a way to **separate signals by their frequency**.

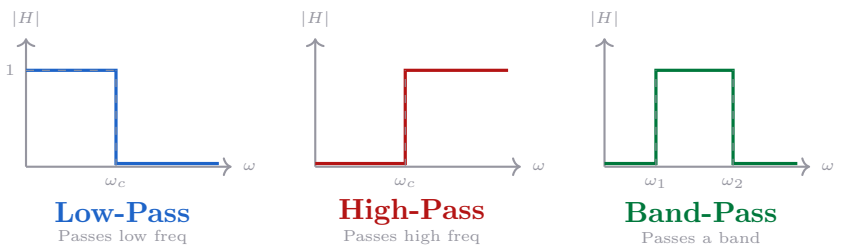
What is Frequency?

Every signal can be thought of as a mixture of **sine waves** at different **frequencies** (how fast they oscillate) and **amplitudes** (how large they are). Frequency is measured in Hertz (Hz) or radians per second (rad/s).

- **Low frequency** = slow oscillation (e.g., bass in music, DC motor speed)
- **High frequency** = fast oscillation (e.g., treble in music, electrical noise)
- $\omega = 2\pi f$ converts Hz to rad/s

1.2 Types of Filters

A filter is a circuit or algorithm that **passes some frequencies and blocks others**.



These are **ideal** (brick-wall) responses. Real filters have a smooth transition. The Butterworth filter is the best known example of a *smooth, practical* low-pass filter.

2 Introducing the Butterworth Filter

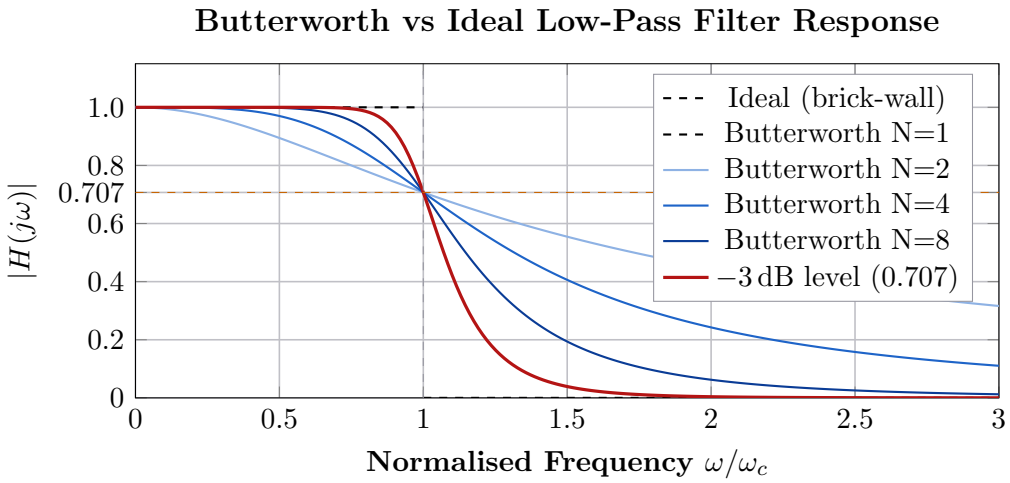
2.1 Who Invented It and Why?

The Butterworth filter was described by British engineer **Stephen Butterworth** in 1930. His goal was to design a filter whose passband gain is as *flat* as possible — no ripple, no bumps. He called this characteristic **maximally flat**.

Maximally Flat — What Does It Mean?

- In the **passband** ($\omega \ll \omega_c$): gain stays very close to 1 (flat, no ripple)
- At the **cutoff** ($\omega = \omega_c$): gain drops to exactly $\frac{1}{\sqrt{2}} \approx 0.707$
- In the **stopband** ($\omega \gg \omega_c$): gain rolls off smoothly at $-20N$ dB/decade
- **No equiripple**, no overshoot — just a gentle, monotone rolloff

2.2 Comparing Filter Types



Key observation: Higher N gives a sharper cutoff and closer approach to the ideal, but requires more components.

3 The Transfer Function and Magnitude Response

3.1 What is a Transfer Function?

Before we define the Butterworth filter mathematically, we need to understand what a transfer function is.

Transfer Function $H(s)$

A **transfer function** describes how a system responds to an input signal. It is a ratio:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\text{Output (in Laplace domain)}}{\text{Input (in Laplace domain)}}$$

where $s = \sigma + j\omega$ is a **complex frequency variable**.

To find the **frequency response** of the filter (how it behaves for pure sinusoids), we substitute $s = j\omega$:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

The **magnitude** $|H(j\omega)|$ tells us how much each frequency is amplified or attenuated.

3.2 Deriving the Butterworth Magnitude Response

Butterworth started with one requirement: the squared magnitude response must be a **rational function of** ω^2 with no ripple in the passband.

The simplest such function that equals 1 at $\omega = 0$ and rolls off monotonically is:

Butterworth Squared Magnitude Response

$$\boxed{|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}}}$$

$$\begin{aligned} N &= \text{filter order (positive integer: 1, 2, 3, ...)} \\ \omega_c &= \text{cutoff frequency in rad/s} \\ \omega &= \text{input signal frequency in rad/s} \end{aligned}$$

3.2.1 Why this formula?

Let's check what happens at key points:

Condition	ω/ω_c	$\lvert H\rvert^2$	$\lvert H\rvert$
DC (zero freq.)	0	$\frac{1}{1+0} = 1$	1 (full pass)
Cutoff	1	$\frac{1}{1+1} = \frac{1}{2}$	$\frac{1}{\sqrt{2}} \approx 0.707$ (-3 dB)
Deep stopband	$\gg 1$	$\approx \frac{1}{(\omega/\omega_c)^{2N}} \to 0$	$\to 0$ (blocked)

3.3 Decibels and the -3 dB Point

Understanding Decibels (dB)

Gain in decibels is defined as:

$$A_{dB} = 20 \log_{10} |H(j\omega)|$$

At the cutoff frequency $\omega = \omega_c$:

$$A_{dB} = 20 \log_{10} \left(\frac{1}{\sqrt{2}} \right) = 20 \times (-0.15051) \approx -3\,\text{dB}$$

This is why ω_c is called the -3 **dB frequency** or **half-power frequency** (since power $\propto |H|^2 = 1/2$).

3.4 Roll-off Rate

For $\omega \gg \omega_c$, the magnitude becomes:

$$|H(j\omega)| \approx \left(\frac{\omega_c}{\omega} \right)^N$$

In dB: every time ω doubles (one octave), gain drops by $6N$ dB. Every tenfold increase (one decade): $-20N$ dB.

Roll-off Rate

$$\text{Roll-off} = -20N \text{ dB/decade} \quad \text{or equivalently} \quad -6N \text{ dB/octave}$$

Example: $N = 4 \Rightarrow -80$ dB/decade. The signal is reduced to 1/10,000 of its input for every tenfold increase in frequency beyond ω_c.

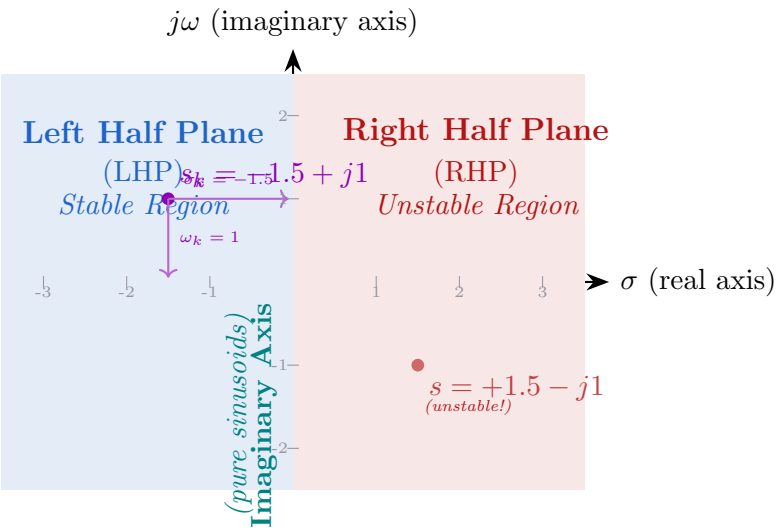
4 The Complex Frequency Plane (s-plane) and Poles

This section is crucial — everything in filter design happens in the s-plane.

4.1 What is the s-plane?

The s-plane is simply a 2D graph where:

- The **horizontal axis** (σ) is the **real part** of s
- The **vertical axis** ($j\omega$) is the **imaginary part** of s
- The imaginary axis ($\sigma = 0$) corresponds to real sinusoidal signals



4.2 What is a Pole?

A **pole** of the transfer function $H(s)$ is a value of s at which $H(s) \to \infty$. Mathematically, poles are the *roots of the denominator* of $H(s)$.

Physical Meaning of a Pole

If a system has a pole at $s_k = \sigma_k + j\omega_k$, then its time-domain response contains a term:

$$c_k \cdot e^{s_k t} = c_k \cdot e^{\sigma_k t} \cdot e^{j\omega_k t}$$

- $e^{j\omega_k t}$ = oscillation at frequency ω_k (the imaginary part)
- $e^{\sigma_k t}$ = envelope that grows or decays (the real part)
- If $\sigma_k < 0$ (LHP): $e^{\sigma_k t} \to 0$ as $t \to \infty$ **STABLE**
- If $\sigma_k > 0$ (RHP): $e^{\sigma_k t} \to \infty$ as $t \to \infty$ **UNSTABLE**
- If $\sigma_k = 0$ (on imaginary axis): oscillates forever (marginally stable)

Bottom line: For a stable (and useful) filter, ALL poles must be in the left half plane.

5 Finding the Butterworth Poles — Full Derivation

5.1 Step 1: From Magnitude to Transfer Function

We have $|H(j\omega)|^2$. To get $H(s)$, we use the **analytic continuation**: replace $j\omega$ with s:

$$H(s)\,H(-s) = \frac{1}{1 + \left(\dfrac{s}{j\omega_c}\right)^{2N}}$$

Since $(j)^{2N} = (j^2)^N = (-1)^N$:

Denominator Equation

$$H(s)\,H(-s) = \frac{\omega_c^{2N}}{\omega_c^{2N} + (-1)^N s^{2N}}$$

The $2N$ poles of $H(s)H(-s)$ are roots of the denominator:

$$\omega_c^{2N} + (-1)^N s^{2N} = 0$$

5.2 Step 2: Solve for the $2N$ Roots

Rearrange:

$$s^{2N} = -(-1)^N \omega_c^{2N} = (-1)^{N+1} \omega_c^{2N}$$

Write $(-1)^{N+1}$ using Euler's formula. Note that $-1 = e^{j\pi}$, so $(-1)^{N+1} = e^{j\pi(N+1)}$. Thus:

$$s^{2N} = \omega_c^{2N} \cdot e^{j\pi(N+1)}$$

Taking the $2N$-th root (with all $2N$ solutions spaced $\frac{2\pi}{2N} = \frac{\pi}{N}$ apart):

Complete Pole Formula — All $2N$ poles

$$\boxed{s_k = \omega_c \exp\left[j\left(\frac{\pi(N+1)}{2N} + \frac{(k-1)\pi}{N}\right)\right], \quad k = 1, 2, \ldots, 2N}$$

Equivalently (the most common textbook form), defining θ_k as the angle:

$$\theta_k = \frac{\pi}{2} + \frac{(2k-1)\pi}{2N} = 90 + \frac{(2k-1) \times 180}{2N}$$

$$s_k = \omega_c\left(\cos\theta_k + j\sin\theta_k\right), \quad k = 1, 2, \ldots, 2N$$

5.3 Step 3: Understand the Geometry

All $2N$ poles lie **on a circle of radius** ω_c, equally spaced by $\frac{180}{N}$.

Why a Circle?

Because $|s_k| = \omega_c$ for all k — every pole has the same magnitude ω_c. This is what "all poles on a circle of radius ω_c" means.

5.4 Step 4: Select Only the LHP Poles

Out of the $2N$ poles, exactly N lie in the left half plane ($90 < \theta < 270$). We **use only these N poles** to form $H(s)$. The other N poles (in the RHP) belong to $H(-s)$ and are discarded.

Why Discard the RHP Poles?

- RHP poles have $\sigma > 0$, giving $e^{\sigma t}$ that grows without bound
- This means the filter's output would explode to infinity for any input — physically impossible
- **Only LHP poles give a stable, causal, physically realizable filter**

5.5 Step 5: Why Are There No Poles at $\pm j\omega_c$?

The pole angles are $\theta_k = 90 + \frac{(2k-1)\times 180}{2N}$. For a pole to land at exactly 90 (i.e., $+j\omega_c$):

$$\frac{(2k-1)\times 180}{2N} = 0 \;\Rightarrow\; 2k-1=0 \;\Rightarrow\; k = 0.5$$

But k must be a positive integer! So no pole ever lands at ± 90.

The Physical Reason

A pole exactly on the imaginary axis would mean the filter oscillates forever — it would be an *oscillator*, not a filter. The Butterworth formula is specifically constructed to avoid this. The $(2k-1)$ factor (always odd) guarantees a $\frac{90}{N}$ offset, keeping poles away from 0, 90, 180, and 270.

6 The Pole Diagram — Fully Annotated

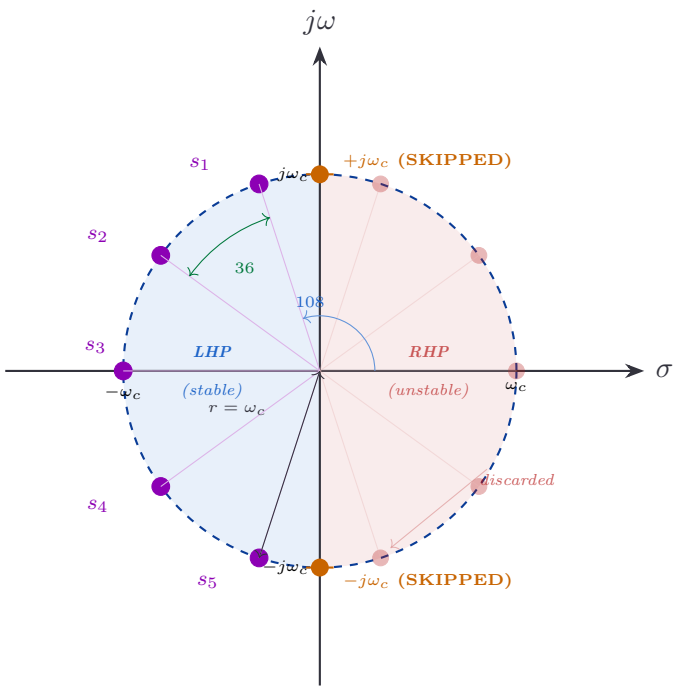


Figure: Pole circle for $N = 5$ (10 poles total). Purple = 5 used LHP poles. Faint red = 5 discarded RHP poles. Orange × = the $\pm j\omega_c$ positions that are always skipped.

7 Building the Transfer Function from Poles

Once we have the N poles, the transfer function is:

Transfer Function from Poles

$$H(s) = \frac{\omega_c^N}{\prod_{k=1}^{N}(s - s_k)}$$

The numerator ω_c^N ensures $H(0) = 1$ (unity gain at DC).

7.1 Complex Conjugate Pairs

Butterworth poles come in complex conjugate pairs (one above and one below the real axis). Multiplying each pair together gives a **real-coefficient quadratic**:

$$(s - s_k)(s - s_k^*) = s^2 - 2\mathrm{Re}(s_k) \cdot s + |s_k|^2 = s^2 - 2\sigma_k s + \omega_c^2$$

Since $\sigma_k = \omega_c \cos\theta_k < 0$ for LHP poles, we write $-2\sigma_k = 2\omega_c|\cos\theta_k| > 0$.

The coefficient $2|\cos\theta_k|$ is called $\frac{\omega_c}{Q_k}$ where Q_k is the **quality factor** of that pole pair.

7.2 Odd vs Even Order

- **Even** N: all poles are in conjugate pairs $\Rightarrow$ $H(s)$ is a product of $N/2$ quadratics
- **Odd** N: one pole is real (at $s = -\omega_c$, i.e., $\theta = 180$), the rest are pairs $\Rightarrow$ $H(s)$ has one $(s + \omega_c)$ factor plus $(N-1)/2$ quadratics

8 Standard Butterworth Polynomial Tables

For normalised filters ($\omega_c = 1$ rad/s), the pole positions and quadratic factors are well-known. They are usually listed as the **Butterworth polynomial** $B_N(s)$ where $H(s) = \frac{1}{B_N(s)}$.

Order N	**Butterworth Polynomial** $B_N(s)$
1	$(s+1)$
2	$s^2 + 1.4142\,s + 1$
3	$(s+1)(s^2+s+1)$
4	$(s^2 + 0.7654\,s + 1)(s^2 + 1.8478\,s + 1)$
5	$(s+1)(s^2 + 0.6180\,s + 1)(s^2 + 1.6180\,s + 1)$
6	$(s^2 + 0.5176\,s + 1)(s^2 + 1.4142\,s + 1)(s^2 + 1.9319\,s + 1)$
7	$(s+1)(s^2 + 0.4450\,s + 1)(s^2 + 1.2470\,s + 1)(s^2 + 1.8019\,s + 1)$
8	$(s^2 + 0.3902\,s + 1)(s^2 + 1.1111\,s + 1)(s^2 + 1.6629\,s + 1)(s^2 + 1.9616\,s + 1)$

Notice: All coefficients are positive $\Rightarrow$ all poles in LHP $\Rightarrow$ always stable.

9 Complete Design Procedure

Here is the full step-by-step method for designing any Butterworth low-pass filter:

Step 1 — Determine filter order N

Given: passband edge ω_p (desired flat region), stopband edge ω_s, max passband loss A_p dB, min stopband attenuation A_s dB.

$$N \geq \frac{\log\left[\sqrt{(10^{A_s/10}-1)/(10^{A_p/10}-1)}\right]}{\log(\omega_s/\omega_p)}$$

Round N **up** to the nearest integer.

Step 2 — Find the exact cutoff frequency

With $A_p = 3\,\text{dB}$: $\omega_c = \omega_p$ directly. Otherwise:

$$\omega_c = \frac{\omega_p}{(10^{A_p/10}-1)^{1/(2N)}}$$

Step 3 — Compute pole angles

$$\theta_k = 90 + \frac{(2k-1)\times 180}{2N}, \quad k = 1, 2, \ldots, N$$

Step 4 — Compute pole positions

$$s_k = \omega_c \cos\theta_k + j\,\omega_c \sin\theta_k$$

Step 5 — Form the transfer function

For each conjugate pair $(s_k,\, s_k^*)$:

$$(s - s_k)(s - s_k^*) = s^2 + 2\omega_c|\cos\theta_k|\, s + \omega_c^2$$

For the real pole at $\theta = 180$ (odd N): factor is $(s + \omega_c)$.
Multiply all factors and put ω_c^N in the numerator.

Step 6 (optional) — Frequency scaling

The design above gives a normalised filter (ω_c as computed). If you need it at a different frequency ω_c', replace $s \to s\,(\omega_c/\omega_c')$.

10 Worked Examples — Orders 1 through 6

10.1 Example 1: First-Order Butterworth ($N = 1$, $\omega_c = 1$)

Pole angle: $\theta_1 = 90 + \frac{1\times 180}{2} = 90 + 90 = 180$

Pole: $s_1 = 1 \cdot \cos 180 + j \cdot 1 \cdot \sin 180 = -1 + j\,0 = -1$

Result: 1st Order

$$H(s) = \frac{1}{s+1}$$

Simple RC low-pass filter with $\omega_c = 1$ rad/s. Roll-off: -20 dB/decade.

10.2 Example 2: Second-Order Butterworth ($N = 2$, $\omega_c = 1$)

Pole angles:

$$\theta_1 = 90 + \frac{1 \times 180}{4} = 90 + 45 = 135$$
$$\theta_2 = 90 + \frac{3 \times 180}{4} = 90 + 135 = 225$$

Poles:

$$s_1 = \cos 135 + j \sin 135 = -\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2} \approx -0.7071 + j\,0.7071$$
$$s_2 = \cos 225 + j \sin 225 = -\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} \approx -0.7071 - j\,0.7071$$

Since s_1 and s_2 are conjugate pairs:

$$(s - s_1)(s - s_2) = s^2 + \sqrt{2}\,s + 1 = s^2 + 1.4142\,s + 1$$

Result: 2nd Order

$$H(s) = \frac{1}{s^2 + 1.4142\,s + 1}$$

Roll-off: -40 dB/decade.

10.3 Example 3: Third-Order Butterworth ($N = 3$, $\omega_c = 1$)

Pole angles:

$$\theta_1 = 90 + \frac{1 \times 180}{6} = 90 + 30 = 120$$
$$\theta_2 = 90 + \frac{3 \times 180}{6} = 90 + 90 = 180 \quad \leftarrow \text{real pole!}$$
$$\theta_3 = 90 + \frac{5 \times 180}{6} = 90 + 150 = 240$$

Poles:

$$s_1 = \cos 120 + j \sin 120 = -0.5 + j\,0.8660$$
$$s_2 = \cos 180 + j \sin 180 = -1 \quad \text{(real pole)}$$
$$s_3 = \cos 240 + j \sin 240 = -0.5 - j\,0.8660$$

(s_1, s_3) form a pair: $(s - s_1)(s - s_3) = s^2 + s + 1$

Result: 3rd Order

$$H(s) = \frac{1}{(s+1)(s^2 + s + 1)}$$

Roll-off: -60 dB/decade.

10.4 Example 4: Fourth-Order Butterworth ($N = 4$, $\omega_c = 1$)

Spacing: $\frac{180}{4} = 45$

Pole angles: $\theta_k = 90 + (2k - 1) \times 22.5$

k	θ_k	$\cos\theta_k$	$\sin\theta_k$
1	112.5	-0.3827	$+0.9239$
2	157.5	-0.9239	$+0.3827$
3	202.5	-0.9239	-0.3827
4	247.5	-0.3827	-0.9239

Pairs: (s_1, s_4) and (s_2, s_3):

$$(s_1, s_4): \; s^2 + 2(0.3827)s + 1 = s^2 + 0.7654\,s + 1$$

$$(s_2, s_3): \; s^2 + 2(0.9239)s + 1 = s^2 + 1.8478\,s + 1$$

Result: 4th Order

$$H(s) = \frac{1}{(s^2 + 0.7654\,s + 1)(s^2 + 1.8478\,s + 1)}$$

Roll-off: -80 dB/decade.

10.5 Example 5: Fifth-Order Butterworth ($N = 5$, $\omega_c = 1$)

Spacing: $\frac{180}{5} = 36$

Pole angles: $90 + 18 = 108$, $90 + 54 = 144$, **180** (real), 216, 252

k	θ_k	s_k	Type
1	108	$-0.3090 + j\,0.9511$	conjugate pair with $k = 5$
2	144	$-0.8090 + j\,0.5878$	conjugate pair with $k = 4$
3	180	-1.0	real pole
4	216	$-0.8090 - j\,0.5878$	pair with $k = 2$
5	252	$-0.3090 - j\,0.9511$	pair with $k = 1$

Result: 5th Order

$$H(s) = \frac{1}{(s + 1)(s^2 + 0.6180\,s + 1)(s^2 + 1.6180\,s + 1)}$$

Roll-off: -100 dB/decade. Note: $0.6180 = 2\cos 72 = 2\sin 18$ and $1.6180 = 2\cos 36 =$ golden ratio!

10.6 Example 6: Sixth-Order Butterworth ($N = 6$, $\omega_c = 1$) — Full Solution

Problem

Design a 6th-order Butterworth low-pass filter with $\omega_c = 1$ rad/s. Find all poles and the transfer function.

Spacing: $\frac{180}{6} = 30$. **Starting angle:** $90 + 15 = 105$.

10.6.1 Pole Angle Table

k	$\theta_k = 90 + (2k-1) \times 15$	$\cos\theta_k$	$\sin\theta_k$	s_k
1	105	-0.2588	$+0.9659$	$-0.2588 + j\,0.9659$
2	135	-0.7071	$+0.7071$	$-0.7071 + j\,0.7071$
3	165	-0.9659	$+0.2588$	$-0.9659 + j\,0.2588$
4	195	-0.9659	-0.2588	$-0.9659 - j\,0.2588$
5	225	-0.7071	-0.7071	$-0.7071 - j\,0.7071$
6	255	-0.2588	-0.9659	$-0.2588 - j\,0.9659$

10.6.2 Conjugate Pairs

Pair	Poles	$-2\cos\theta_k$	Quadratic Factor
(s_1, s_6)	$-0.2588 \pm j\,0.9659$	$2 \times 0.2588 = 0.5176$	$s^2 + 0.5176\,s + 1$
(s_2, s_5)	$-0.7071 \pm j\,0.7071$	$2 \times 0.7071 = 1.4142$	$s^2 + 1.4142\,s + 1$
(s_3, s_4)	$-0.9659 \pm j\,0.2588$	$2 \times 0.9659 = 1.9319$	$s^2 + 1.9319\,s + 1$

10.6.3 Pole Diagram

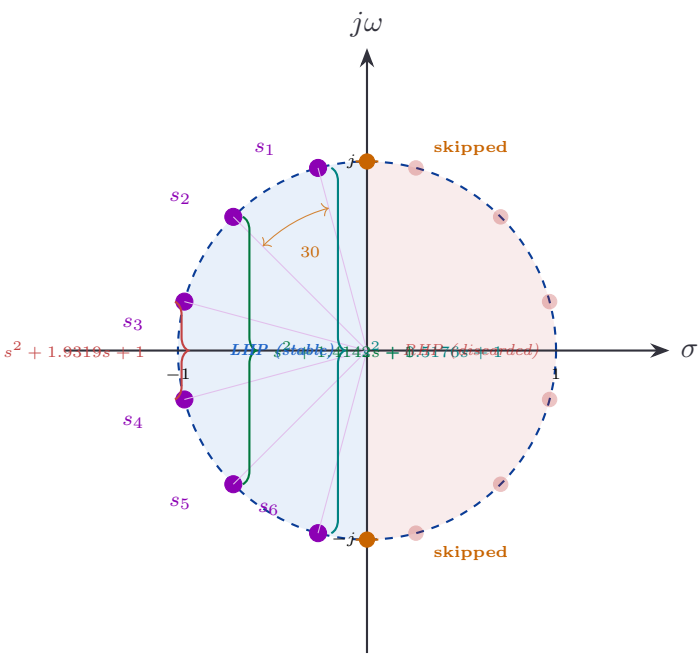


Figure: All 6 poles for 6th-order Butterworth ($\omega_c = 1$). Braces show conjugate pairs and their quadratic factors.

10.6.4 Final Transfer Function

6th-Order Butterworth, $\omega_c = 1$ rad/s

$$\boxed{H(s) = \frac{1}{(s^2 + 0.5176\,s + 1)(s^2 + 1.4142\,s + 1)(s^2 + 1.9319\,s + 1)}}$$

Roll-off: -120 dB/decade.
Verification: At $s = j\omega$, $|H(j\cdot 1)|$ should $= 1/\sqrt{2}$. Each factor at $s = j$: $|j^2 + bj + 1| = |bj| = b$. So $|H(j)| = 1/(0.5176 \times 1.4142 \times 1.9319) = 1/1.414 = 1/\sqrt{2}$ ✓

11 Frequency Scaling and Denormalisation

The transfer functions above use **normalised** $\omega_c = 1$ rad/s. To get a filter at any other cutoff Ω_c:

Frequency Scaling Rule

Replace every s with $\dfrac{s}{\Omega_c}$ in the normalised transfer function.

For a quadratic factor $s^2 + b\,s + 1$ at $\omega_c = 1$:

$$\left(\frac{s}{\Omega_c}\right)^2 + b\left(\frac{s}{\Omega_c}\right) + 1 \implies s^2 + b\,\Omega_c\,s + \Omega_c^2$$

In general: multiply the s coefficient by Ω_c and the constant term by Ω_c^2.

Quick Example: Scale the 2nd-order filter to $\Omega_c = 1000$ rad/s:

$$H_{\text{norm}}(s) = \frac{1}{s^2 + 1.4142s + 1} \xrightarrow{s \to s/1000} H(s) = \frac{1}{\left(\frac{s}{1000}\right)^2 + 1.4142\left(\frac{s}{1000}\right) + 1} = \frac{10^6}{s^2 + 1414.2\,s + 10^6}$$

12 Key Properties Summary

Butterworth Filter — Complete Properties Reference

Magnitude response	$\lvert H(j\omega)\rvert^2 = \dfrac{1}{1+(\omega/\omega_c)^{2N}}$
Cutoff condition	$\lvert H(j\omega_c)\rvert = 1/\sqrt{2} \approx 0.707$ ($-3\,\text{dB}$)
Passband	Maximally flat — no ripple
Roll-off rate	$-20N$ dB/decade beyond ω_c
Pole locations	Circle of radius ω_c, equally spaced $180/N$
Pole angles	$\theta_k = 90 + (2k-1)\cdot 90/N$, $k = 1\ldots N$
LHP poles only	$\theta_k \in (90, 270)$ — ensures stability
Imaginary axis	Never has poles — always avoids ± 90
Pole symmetry	Complex conjugate pairs; real pole only for odd N (at 180)
Minimum phase	Yes — all zeros at $s = \infty$
Phase response	Not linear (unlike Bessel filter)
Compared to Chebyshev	Butterworth has no passband ripple but slower rolloff
Compared to Elliptic	Butterworth is monotone; elliptic has steeper cutoff but ripple

13 Common Mistakes and How to Avoid Them

Mistake 1 — Using All $2N$ Poles

After computing $2N$ roots, beginners sometimes use all of them. **You must use only the N LHP poles.** The other N are the poles of $H(-s)$, not $H(s)$. Using RHP poles will give an unstable filter.

Mistake 2 — Forgetting the Numerator

The transfer function is $H(s) = \dfrac{\omega_c^N}{\prod(s-s_k)}$, **not** $\dfrac{1}{\prod(s-s_k)}$. The ω_c^N in the numerator ensures DC gain $= 1$. For normalised filters ($\omega_c = 1$), this is just 1, so it's easy to forget for unnormalised filters.

Mistake 3 — Wrong Sign in Quadratic

The quadratic factor from pole pair $s_k = \sigma_k \pm j\omega_k$ is:

$$s^2 - 2\sigma_k s + \omega_c^2$$

Since $\sigma_k < 0$ for LHP poles, $-2\sigma_k > 0$. The coefficient of s is **positive**. If you get a negative coefficient, you have used the wrong sign somewhere.

Mistake 4 — Mixing Radians and Degrees

The pole angle formula gives angles in degrees. When using a calculator for cos and sin, ensure you are in **degree mode**, or convert: $\theta_{\text{rad}} = \theta_{\text{deg}} \times \pi/180$.

14 Quick Reference Card

Butterworth Filter — Quick Reference

Magnitude:	$\lvert H\rvert^2 = \dfrac{1}{1+(\omega/\omega_c)^{2N}}$
Pole count:	Use N poles (LHP only)
Pole angle:	$\theta_k = 90 + \dfrac{(2k-1)\times 180}{2N}, \quad k = 1\ldots N$
Pole position:	$s_k = \omega_c(\cos\theta_k + j\sin\theta_k)$
Spacing:	$\Delta\theta = 180/N$ between adjacent poles
Quadratic:	$(s-s_k)(s-s_k^*) = s^2 + 2\omega_c\lvert\cos\theta_k\rvert\, s + \omega_c^2$
Real pole:	Only for odd N, at $\theta = 180$: factor $(s+\omega_c)$
Roll-off:	$-20N$ dB/decade
Stability:	All $\theta_k \in (90, 270) \Rightarrow$ always stable
Scale to Ω_c:	Replace $s \to s/\Omega_c$ in normalised $H(s)$

The 3-Sentence Summary

1. A Butterworth filter has a perfectly flat passband and rolls off at $-20N$ dB/decade past the cutoff ω_c.
2. Its poles lie equally spaced on a circle of radius ω_c, and we keep only the N poles in the left half of the s-plane (for stability), always offset so no pole falls on the imaginary axis.
3. Once you have the poles, group them into real quadratic factors and write $H(s)$ as the ratio of ω_c^N over their product.